

Antibandwidth of Bipartite Graphs

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Abstract

Let $G = (V, E)$ be a graph with n vertices and $f : V(G) \rightarrow \{1, 2, \dots, n\}$ be a bijective function. We define the minimum edge difference as $ab(G, f) = \min\{|f(u) - f(v)| : (u, v) \in E\}$. We say that f is a k -antibandwidth labeling on G if $ab(G, f) \geq k$. Let $ab(G) = \max_f\{ab(G, f)\}$. We therefore investigate the lower bound of $ab(G)$. In other words, to what extent can we maximize the minimum edge difference in G among all possible labelings? We first give a lower bound $\lfloor n/3\Delta(G) \rfloor$ on $ab(G, f)$. In addition, we conjecture that for any graph G with $n > 2 \cdot \Delta(G)$, we have $ab(G) \geq \lfloor n/\Delta(G) \rfloor$, where $\Delta(G)$ is the maximum degree of G , and show that our conjecture holds for all connected bipartite graphs with more than $2 \cdot \Delta(G)$ vertices. Besides solving the problem of finding the maximized minimum-difference labeling, we further explore the relationships between the antibandwidth problems and the equitable coloring problems.

Keywords: antibandwidth, bipartite graph, equitable coloring.

1 Introduction

We study a labeling method called *antibandwidth labeling*. Let $G = (V, E)$ be a graph with n vertices, and $\deg(v)$ the degree of a vertex $v \in V$ and let $\Delta(G)$ be the maximum degree of G . Let $f : V(G) \rightarrow \{1, 2, \dots, n\}$ be a bijective function. The *minimum difference* of f on G , denoted by $ab(G, f)$, is defined as $\min\{|f(u) - f(v)| : (u, v) \in E\}$. We say that f is a k -antibandwidth labeling on G if $ab(G, f) \geq k$. Figure 1 shows an example of a 3-antibandwidth labeling of the Petersen graph. Let $ab(G) = \max_f\{ab(G, f)\}$ denote the maximized minimum-difference among all antibandwidth labelings of G . We abbreviate $\Delta(G)$ to Δ when no ambiguity arises.

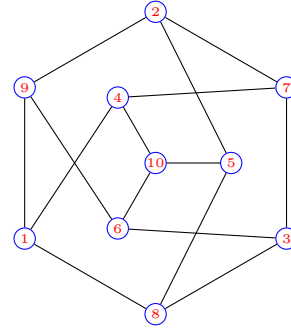


Figure 1: An antibandwidth labeling of the Petersen graph.

Many other labeling problems origin from the *channel assignment problem*. In a channel assignment problem, each vertex in G corresponds to a transmitter location with the label assigned to it to determine its frequency, and each edge connecting two locations indicates that they are within a distance from each other. In such a communication system, we try to find a frequency assignment method which guarantees interference avoidance. In the antibandwidth labeling problem, we further require that every vertex in G should be as isolated as possible, so that the noise between any two locations can be minimized.

It is known that the antibandwidth problem is NP-complete [8]. There are also exact results for some special graphs, such as paths, cycles, special trees, and complete bipartite graphs [10, 12, 16]; recently, the antibandwidth of three-dimensional meshes is also been investigated [15]. In the previous results [10], it is shown that for any bipartite graph G , $ab(G) \geq \lceil D(G)/2 \rceil$, where $D(G)$ denotes the diameter of G . In section 5, we will give another lower bound for the antibandwidth of all connected bipartite graphs with $n > 2\Delta$ vertices, which indeed matches the upper bound for some cases.

In contrast with maximizing the minimum edge difference, there is a dual problem called the *graph*

bandwidth problem. The *bandwidth* of a graph G is the minimum over all one-to-one labelings $f : V \rightarrow \{1, 2, \dots, n\}$ of G of the largest edge difference, that is, the minimum of $\max\{|f(u) - f(v)| : (u, v) \in E\}$ among all feasible labeling. Motivated by sparse matrix computation, this problem has been extensively studied (e.g., [2, 11]). Papadimitriou proved that the problem of determining the bandwidth of a graph is NP-complete [14]; later on, Garey *et al.* [4] showed that it is NP-complete to determine the bandwidth even when G is restricted to trees of maximum degree 3. Furthermore, it was known that it is NP-hard to approximate the bandwidth within a factor better than $3/2$, and it is NP-hard to approximate the bandwidth within any constant even for some certain kind of graphs [2].

2 Antibandwidth Labelings vs. Equitable Colorings

The antibandwidth problem can be viewed as a robust version of the *equitable coloring*. An equitable coloring is an assignment of colors to the vertices of an undirected graph, in such a way that no two adjacent vertices have the same color, and the numbers of vertices in any two color classes differ by at most one.

Paul Erdős [3] (1964) conjectured that any graph with maximum degree Δ has an equitable coloring with $\Delta + 1$ colors. This had been proven by Hajnal and Szemerédi [5], which is known as the Hajnal-Szemerédi theorem (1970). Later, Chen *et al.* [1] conjectured that if G is not a complete graph, an odd cycle, or a complete bipartite graph $K_{2r+1, 2r+1}$ for any $r \geq 1$, then G is equitably Δ -colorable. Kierstead *et al.* [6] shows that for any graph G , an equitable coloring with $\Delta + 1$ colors can be obtained in $O(\Delta n^2)$ time (2010). The smallest integer k for which G is equitably k -colorable is defined as the *equitable chromatic number* of G , denoted by $\chi_{\text{eq}}(G)$. The problem of deciding if $\chi_{\text{eq}}(G) \leq 3$ is NP-complete [7]. It is known that every tree T can be equitably colored with $\lceil \Delta(T)/2 \rceil + 1$ colors [13], and for any connected bipartite graph G which is not a complete bipartite, $\chi_{\text{eq}}(G) \leq \Delta$ [9].

Given a antibandwidth labeling of a graph G , since we can guarantee that no two vertices having the label numbers within a certain range are adjacent to each other, it is straightforward to have an equitable coloring of G by equally and sequentially partitioning the vertex set according to their labels

(see Theorem 1 and Corollary 1). Therefore, we may say that the antibandwidth labelings is more generalized than the equitable colorings from such a point of view. Here we show how the maximized minimum-difference labeling guarantees a way to uniformly separate the labels.

Theorem 1. *If there is a k -antibandwidth labeling of G , then there exists a $(\lfloor \frac{n}{k} \rfloor + 1)$ -equitable coloring on G .*

Proof. Let $r = \lfloor n/k \rfloor + 1$. Given a k -antibandwidth labeling f on G . Define an equitable coloring $g : V \rightarrow \{c_1, c_2, \dots, c_r\}$ by $g(v_i) = c_j$, where $j = \lfloor \frac{f(i)-1}{k} \rfloor + 1$, for $i = 1, 2, \dots, n$. The number of vertices in every color set would differ by at most 1. Since for any two vertices v_i, v_j with $g(v_i) = g(v_j) = c_l$, for some $l \in \{1, 2, \dots, r\}$, we have $k(l-1) \leq f(i) \leq kl$ and $k(l-1) \leq f(j) \leq kl$, then $|f(i) - f(j)| < k$ and thus $(v_i, v_j) \notin E$. Therefore, g is an r -equitable coloring on G . $\square$

Theorem 1 shows how to partition the vertices of G based on their labels to form an equitable coloring. With a similar argument, we have the following corollary.

Corollary 1. *If there is a $\lfloor n/k \rfloor$ -antibandwidth labeling of G , then there exists an r -equitable coloring on G , for all $r \geq k$.*

How about the other way around? We then ask whether a lower bound of $\text{ab}(G)$ can be deduced when having some information about the equitable colorings on G . For example, we use 4 colors $\{a, b, c, d\}$ in Figure 2 to equitably color a graph G with $n = 20$ vertices. By assigning a level range to every color set and numbering those vertices in a color set, we wonder whether there exists a transformation from an equitable coloring to an ϵ -antibandwidth labeling, for some ϵ .

3 Lower Bounds and Upper Bounds of Antibandwidth

Relating the results of equitable colorings to antibandwidth labelings, we guess that there is a lower bound on the maximum-minimum difference for any graph G with $n > 2\Delta$, where n is the number of vertices in it. We have here the constraint $n > 2\Delta$ for the reason that there are some trivial cases on certain graphs. For instance, it is obvious that $\text{ab}(G) = 1$ for any complete graph or any star graph; another example is the complete bipartite graphs $K_{m,n}$, which has the same lower

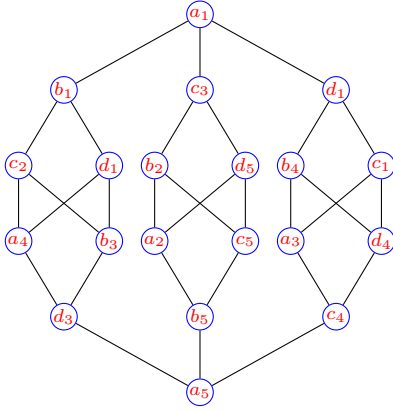


Figure 2: Transforming an equitable coloring with k colors to a $\lfloor n/k \rfloor$ -antibandwidth labeling.

bound $\text{ab}(G) = 1$. Hence, we propose the following conjecture.

Conjecture 1. *For any graph G with $n > 2\Delta$ vertices, $\text{ab}(G) \geq \lfloor \frac{n}{\Delta} \rfloor$.*

Note that in [10], Lin and Yuan give an upper bound $\text{ab}(G) \leq \lfloor \frac{n-1}{\chi(G)-1} \rfloor$, where $\chi(G)$ is the chromatic number of G , which is similar to the form of our predicted lower bound.

There is also a trivial upper bound on $\text{ab}(G)$ for any connected graph. Suppose that G is a connected graph with m disjoint maximum cliques of size k . Since each single clique covers at least $\text{ab}(G)(k-1) + 1$ labels, and thus m maximum cliques would cover at least $\text{ab}(G)(k-1) + m$ labels, we have $\text{ab}(G) \leq \lfloor (n-m)/(k-1) \rfloor$. Figure 3 shows an example of a 6-antibandwidth labeling on a graph G , where $n = 18$, $m = 6$ and $k = 3$, which matches the upper bound.

4 Antibandwidth Labelings of General Graphs

For general graphs, we show that $\lfloor \frac{n}{3\Delta} \rfloor$ is a lower bound of $\text{ab}(G)$ by proposing a greedy $\lfloor \frac{n}{3\Delta} \rfloor$ -antibandwidth labeling algorithm.

Theorem 2. *For any graph G with n vertices, $\text{ab}(G) \geq \lfloor \frac{n}{3\Delta} \rfloor$.*

Proof. Let $\text{FL}(v, \epsilon) = \{i : |i - f(u)| < \epsilon \text{ for each neighbour } u \text{ of } v\}$, $\text{AL}(v, \epsilon) = \{1, 2, \dots, |V(G)|\} \setminus \text{FL}(v, \epsilon)$, $\text{AV}(i, \epsilon) = \{u : i \in \text{AL}(u, \epsilon)\}$ and $\text{FV}(i, \epsilon) = \{u : i \in \text{FL}(u, \epsilon)\}$. We claim that when $\epsilon \leq \lfloor \frac{|V(G)|}{3\Delta} \rfloor$, each vertex will be labeled with one number by GREEDYANTIBANDWIDTH.

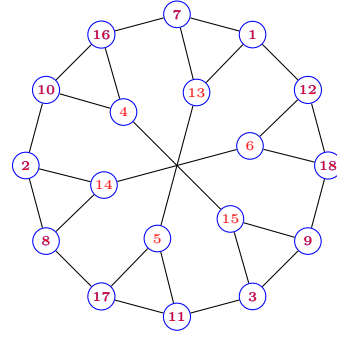


Figure 3: A 6-antibandwidth labeling of a graph with six 3-cliques.

Here we need to show that there is a vertex $u \in \text{AV}(i, \epsilon)$ and an unlabeled vertex v satisfying $f(u) \in \text{AL}(v, \epsilon)$. First, the number of neighbours of vertices which are labeled with $i - (\epsilon - 1)$ to $i - 1$ is at most $(\epsilon - 1)\Delta$. For any vertex v , $|\text{FL}(v, \epsilon)| \leq (2\epsilon - 1)\Delta$. Whenever v cannot be labeled by i , there is a neighbour v' of v labeled with value in the range $[i - \epsilon + 1, i - 1]$. Since we have $n - [(\epsilon - 1)(\Delta + 1) + (2\epsilon - 1)\Delta - (\epsilon - 1)] = n - (3\epsilon - 2)\Delta > 0$, there is a vertex $u \in \text{AV}(i, \epsilon)$ such that $f(u) \in \text{AL}(v, \epsilon)$. $\square$

Based on the idea shown in the proof of Theorem 2, Algorithm 1 gives a $\lfloor \frac{n}{3\Delta} \rfloor$ -antibandwidth labeling of G .

Algorithm 1 GREEDYANTIBANDWIDTH(G, ϵ).

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1: for  $i = 1, \dots, n$  do
2:   if there is an unlabeled vertex  $u$  in  $\text{AV}(i, \epsilon)$  then
3:     Label  $u$  with  $i$ .
4:   else if there is a vertex  $u$  in  $\text{AV}(i, \epsilon)$  with  $f(u) \in \text{AL}(v, \epsilon)$  for some unlabeled vertex  $v$  then
5:     Label  $u$  with  $i$  and label  $v$  with  $f(u)$ .
6: return  $f$ 
    
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5 Antibandwidth of Bipartite Graphs

In this section, we show that for every connected bipartite graph G with $n > 2\Delta$ vertices, $\text{ab}(G) \geq \lfloor n/\Delta \rfloor$.

Lemma 1. *Let $G = (U \cup V, E)$ be a connected bipartite graph with n vertices, where U and V are both independent sets with $|U| = n_1$, $|V| = n_2$ and*

$n_1 \geq n_2$. Suppose that $\deg(v) = \Delta$, for all $v \in V$. If $\Delta(G) \geq 3$ and $n > 2 \cdot \Delta(G)$, then there exists a k -antibandwidth labeling of G , where $k \geq \lfloor \frac{n}{\Delta} \rfloor$.

Proof. Let $\epsilon = \lfloor \frac{n}{\Delta} \rfloor$. Suppose if $n_2 \leq (\epsilon - 1)$. Since every vertex in V has degree Δ , we have $|E| = \Delta n_2$. Then, $n_1 \leq \Delta n_2 - (n_2 - 1)$ since G must be connected, and thus $n_1 + n_2 - 1 \leq \Delta n_2 \leq \Delta(\epsilon - 1) \Rightarrow n_1 + n_2 + (\Delta - 1) \leq \Delta \epsilon \leq n$, a contradiction. Therefore we may assume that $n_2 \geq \epsilon$.

Let $U = \{u_1, u_2, \dots, u_{n_1}\}$ and $V = \{v_1, v_2, \dots, v_{n_2}\}$. Here we want to show that there exists an ϵ -antibandwidth labeling f on G by first labeling the vertices in U using labels in $\{1, 2, \dots, n_1\}$, then the vertices in V the labels in $\{n_1 + 1, n_1 + 2, \dots, n_1 + n_2\}$. Since U and V are both independent sets, in the labeling procedure we only need to use labels from $\{n_1 - \epsilon + 1, n_1 - \epsilon + 2, \dots, n_1\}$ carefully so that any vertex $u \in U$ with $n_1 + 1 - \epsilon < f(u) \leq n_1$ does not adjacent to any vertex v in V where $|f(u) - f(v)| \geq \epsilon$.

Since there are Δn_2 edges in G , there exists a vertex $u_1 \in U$ such that $\deg(u_1) \leq \lfloor \Delta \frac{n_2}{n_1} \rfloor$. Let V_1 denote the set of all vertices in V which are adjacent to u_1 and $a_1 = |V_1| \leq \lfloor \Delta \frac{n_2}{n_1} \rfloor$.

Since there are $\Delta \cdot (n_2 - a_1)$ edges connected to $U - \{u_1\}$ from $V - V_1$, there exists a vertex $u_2 \in U - \{u_1\}$ which connects to no more than $\lfloor \Delta \frac{n_1 - a_1}{n_1 - 1} \rfloor$ vertices in $V - V_1$. Let $a_2 = |V_1 \cap V_2|$, where V_2 is the set of all vertices in V which is adjacent to u_2 , from the above observation we get $a_2 \leq \lfloor \Delta \frac{n_1 - a_1}{n_1 - 1} \rfloor$.

By a similar argument, we can always find a vertex $u_{i+1} \in U$ which connects to at most $\lfloor \Delta \frac{n_1 - s_i}{n_1 - i} \rfloor$ vertices in $V - (V_1 \cup V_2 \cup \dots \cup V_i)$, where $s_i = \sum_{j=1}^i a_j$. Repeat the above procedure until $u_1, u_2, \dots, u_{\epsilon-1}$ are found, we have $a_{i+1} \leq \lfloor \Delta \frac{n_1 - s_i}{n_1 - i} \rfloor$, for $i = 1$ to $\epsilon - 1$.

Next, we claim that $s_i < n_2 - \epsilon + i$, for all i , since this would imply that $s_{\epsilon-1} < n_2 - 1$, which indicates that there is always a vertex left in V which can be labeled by $n_1 - i + \epsilon$ after u_i is determined. In the following, we shall prove this claim to complete the proof of this lemma.

Let $x_i = n_2 - \epsilon + i - s_i$. Then $x_{i+1} = n_2 - \epsilon + i + 1 - s_i - a_{i+1} = x_i + 1 - a_{i+1}$. We next prove by induction on i that $x_i > 0$, for all $1 \leq i \leq \epsilon - 1$. It is trivial for the base case that $n_2 - s_1 - \epsilon + 1 > 0$. Suppose the statement is true so that $x_j > 0$ for all $k \leq i$. Consider the case when $k = i + 1$. If $x_i \geq \Delta - 1$, it is obvious that $x_{i+1} \geq 0$ since $a_{i+1} \leq \Delta$. Otherwise, by the induction hypothesis suppose if $0 < x_i < \Delta - 1$, then we have $a_{i+1} \leq \lfloor \Delta \frac{n_2 - s_i}{n_1 - i} \rfloor =$

$\lfloor \Delta \frac{\epsilon - i + x_i}{n_1 - i} \rfloor = \lfloor \Delta \frac{\epsilon - i}{n_1 - i} + \Delta \frac{x_i}{n_1 - i} \rfloor = \lfloor \alpha + \beta x_i \rfloor$, where $\alpha = \Delta \frac{\epsilon - i}{n_1 - i}$ and $\beta = \frac{\Delta}{n_1 - i}$.

Since $n - \Delta i < n - 2i$, we have $\Delta(\epsilon - i) < 2n_1 - 2i$, and thus $\alpha = \Delta \frac{\epsilon - i}{n_1 - i} < 2$. Since $n_1 - i - \Delta \geq n_1 - (\epsilon - 1) - \Delta = \frac{1}{2}(2n_1 - 2\epsilon - 2\Delta + 2) \geq \frac{1}{2}(\epsilon\Delta - 2\epsilon - 2\Delta + 2) = \frac{1}{2}((\epsilon - 2)(\Delta - 2) - 2)$ and $\Delta \geq 3$, we have $n_1 - i - \Delta \geq 0$ if $\epsilon \geq 4$ or $\Delta \geq 4$. Suppose that $\Delta = 3$. If $\epsilon = 2$, since $2n_1 \geq n \Rightarrow n_1 > \Delta$, we have $n_1 - i - \Delta \geq n_1 - \epsilon + 1 - \Delta > -1$; otherwise if $\epsilon = 3$, since $n \geq 9 \Rightarrow n_1 \geq 5$, we have $n_1 - i - \Delta \geq n_1 - \epsilon + 1 - \Delta = 0$. Hence, we have $\beta \leq 1$ for all cases.

By the fact that $\alpha < 2$ and $\beta \leq 1$, we get $a_{i+1} < x_i + 1$, and thus $x_{i+1} = x_i + 1 - a_{i+1} > 0$. Therefore, we have $s_i < n_2 - \epsilon + i$, for $i = 1, 2, \dots, \epsilon - 1$, which completes the proof. $\square$

Theorem 3. For any connected bipartite graph G with n vertices, where $n > 2\Delta$ and $\Delta \geq 3$, there is a $\lfloor \frac{n}{\Delta} \rfloor$ -antibandwidth labeling of G .

Proof. Given any connected bipartite graph G , we can arbitrarily add edges on it until every vertex in V has degree Δ , and then apply the labeling method mentioned in Lemma 1 to get a $\lfloor \frac{n}{\Delta} \rfloor$ -antibandwidth labeling of G . $\square$

Note that $\lfloor \frac{n}{\Delta} \rfloor$ is in fact a tight bound for maximizing the minimum difference of some bipartite graphs. See Figure 4 for example. It shows a graph $G = (U \cup V, E)$ with 12 vertices, with every vertex in G has degree $\Delta = 3$, and $|U| = |V| = 6$. Let $\epsilon = \lfloor \frac{n}{\Delta} \rfloor = 4$, we shall show that $\text{ab}(G) = \epsilon$. Suppose that there exists a k -antibandwidth labeling f of G , where $k \geq 5$. If there is a vertex $u \in U$ such that $f(u) = 5$, for some antibandwidth labeling f . (The same argument can be applied if there is a vertex $v \in V$ such that $f(v) = 5$.) Then, label 5 forces labels in $\{10, 11, 12\}$ to be settled in V for those vertices which are adjacent to u ; similarly, labels 6, 7, 8 forces labels in $\{1, 11, 12\}, \{1, 2, 12\}, \{1, 2, 3\}$ to connect with them respectively. Therefore, the vertices in V are settled with labels in $\{1, 2, 3, 10, 11, 12\}$. Suppose that v is a vertex in V such that $f(v) = 12$, which is already connected to 3 vertices in U with labels in $\{5, 6, 7\}$. Consider another vertex $u' \in U$ with label 4. It follows that u' cannot connect to vertices with labels in $\{1, 2, 3, 12\}$. However, this would contradict to the fact that $\deg(u') = 3$.

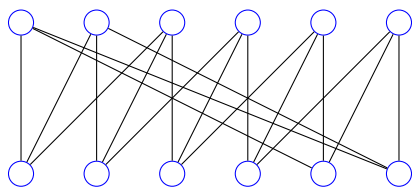


Figure 4: A bipartite graph G with $n = 12$, $\Delta = 3$, and $\text{ab}(G) = 4$.

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